

Benjamin (Ben) Rachunok

915 Partners Way, Rm 4171 – Raleigh, NC 27606

☎ +1 (910) 398-7206 • 📞 +1 (919) 513-7220 • 📠 (919) 515 5281

✉ barachun@ncsu.edu • 🌐 <http://brachunok.com>

Academic Appointments

Assistant Professor

North Carolina State University, Raleigh, NC

Industrial and Systems Engineering

2023–

Graduate Faculty

North Carolina State University, Raleigh, NC

Operations Research

2023–

R&D Intern

Sandia National Labs, Albuquerque, NM

Supervisor: Dr. Andrea Staid

Center for Computing Research

2017–2020

Education & Training

Postdoctoral Researcher

Stanford University, Stanford, CA

Advisor: Dr. Sarah Fletcher

Civil and Environmental Engineering

2020–2022

Ph.D.

Purdue University, West Lafayette, IN

Advisor: Dr. Roshanak Nateghi

Industrial Engineering

2016–2020

Bachelor of Science

North Carolina State University, Raleigh, NC

Industrial and Systems Engineering

2011–2016

Research Support

Collaborations in Artificial Intelligence and Geosciences (CAIG)

\$389,206 NC State / \$884,606 total, Co-PI

Artificial Intelligence-Driven Optimization of Coastal Morphodynamics and Infrastructure Resilience

Collaborative PIs: Katherine Anarde (NC State), Brad Murray (Duke), Laura Moore (UNC-Chapel Hill)

National Science Foundation

August 2026–July 2029

Infrastructure Systems and People

\$80,054 NC State / \$697,903 total, PI

AI-Enabled Decision Support for Rate Setting and Infrastructure Investment in Decentralized Water Utilities under Drought

Collaborative PIs: Marta Zaniolo (Duke), Eric Edwards (UC Davis), Mehdi Nemati (UC Riverside)

National Science Foundation

September 2026–August 2029

North Carolina Collaboratory

\$99,820, Lead PI

North Carolina Data Center Water Use Analysis

State of North Carolina

May 2025–December 2026

North Carolina Collaboratory
\$99,719, *Lead PI*
North Carolina Power System Restoration Optimization

State of North Carolina
May 2025-December 2026

Digitizing Utilities Prize, Phase 1
\$75,000, *Lead PI*
North Carolina Short-term Power Outage Prediction

Department of Energy
June 2024-June 2025

R.L. Rabb Science and Society Fund
\$7625, *Lead PI*
North Carolina Climate & Health Systems Impacts Symposium

North Carolina State University
June 2024-June 2025

Research Collaboration Fund
\$30,000, *Lead NC State PI*
Leveraging AI for Real-time Prediction of Extreme Weather Events
to Foster Climate Resilience: a Global Research Alliance

University Global Partnership Network
August 2024-August 2025

UPS Endowment Fund
\$100,000, *Co-PI*
Assessing Drought Impacts on Urban Water Affordability and Access

Stanford University
September 2022-August 2023

Peer-Reviewed Journal Articles

* indicates equal contributions, Underline indicates an advisee

- [30] Y. Essus, **B. Rachunok**, T. Logan, K. Hardaway, and D. Bristow. "Towards Operationalizing Ecological Resilience for Infrastructure Decision-Making". In: *Risk Analysis* (2026). doi: 10.1111/risa.70316.
- [29] A. Helmrich, J. R. Wolfenbarger, K. Hardaway, **B. Rachunok**, and M. Haraguchi. "Flexible infrastructure design for changing conditions". In: *Environmental Research: Infrastructure and Sustainability* (2026). doi: 10.1088/2634-4505/ae9c6f.
- [28] P. Leavitt, F. Livingston, B. McConnell, and **B. Rachunok**. "Anticipating Household Rescue Demand in Hurricanes Using Socio-Demographic Data and Machine Learning". In: *International Journal of Disaster Risk Reduction* 145 (2026), p. 106405. doi: 10.1016/j.ijdr.2026.106405.
- [27] T. Logan, **B. Rachunok**, K. Hardaway, D. Bristow, A. Reilly, D. Johnson, J. Johansson, G. Cremen, J. Boakye, and P. Schweizer. "Advancing systemic risk assessment for complex, interdependent systems: a research agenda". In: *Risk Analysis* 46.5 (2026). doi: 10.1111/risa.70243.
- [26] Y. Essus and **B. Rachunok**. "A Computational Framework for Modeling Access to Services and the Resilience Impacts of Electric Vehicles During Blackouts". In: *IEEE Access* (2025). doi: 10.1109/ACCESS.2025.3629567.
- [25] K. Hardaway, K. Best, and **B. Rachunok**. "Leveraging the American Housing Survey to Quantify Risk and Resilience". In: *Risk Analysis* (2025). doi: 10.1111/risa.70100.
- [24] Y. Essus and **B. Rachunok**. "Electric Vehicles Limit Equitable Access to Essential Services During Blackouts". In: *NPJ Urban Mobility* (2024). doi: 10.1038/s44333-024-00007-3.
- [23] **B. Rachunok**, A. Verma, and S. Fletcher. "Predicting and understanding residential water use with interpretable machine learning". In: *Environmental Research Letters* 19.1 (2024). doi: 10.1088/1748-9326/ad1434.
- [22] J. Skerer, A. Verma, M. Edwards, **B. Rachunok**, and S. Fletcher. "Alternative household water affordability metrics using water bill delinquency behavior". In: *Environmental Research Letters* (2024). doi: 10.1088/1748-9326/ad5609.

- [21] A. Nayak, **B. Rachunok**, B. Thompson, and S. Fletcher. "Socio-hydrological impacts of rate design on water affordability during drought". In: *Environmental Research Letters* 18.12 (2023). doi: 10.1088/1748-9326/ad0994.
- [20] **B. Rachunok** and S. Fletcher. "Socio-hydrological drought impacts on urban water affordability". In: *Nature Water* 1.1 (2023). doi: 10.1038/s44221-022-00009-w.
- [19] J. Bennett, **B. Rachunok**, R. Flage, and R. Nateghi. "Mapping climate discourse to climate opinion: An approach for augmenting surveys with social media to enhance understandings of climate opinion in the United States". In: *PLOS ONE* (2021). doi: 10.1371/journal.pone.0245319.
- [18] M. Choi, **B. Rachunok**, and R. Nateghi. "Short-term solar irradiance forecasting using convolutional neural networks and cloud imagery". In: *Environmental Research Letters* 16.4 (2021). doi: 10.1088/1748-9326/abe06d.
- [17] R. Obringer, **B. Rachunok**, D. Maia-Silva, M. Arbabzadeh, R. Nateghi, and K. Madani. "The overlooked environmental footprint of increasing Internet use". In: *Resources, Conservation and Recycling* 167 (2021). doi: 10.1016/j.resconrec.2020.105389.
- [16] **B. Rachunok**, J. Bennett, R. Flage, and R. Nateghi. "A path forward for leveraging social media to improve the study of community resilience". In: *International Journal of Disaster Risk Reduction* 59 (2021). doi: 10.1016/j.ijdrr.2021.102236.
- [15] **B. Rachunok** and R. Nateghi. "Overemphasis on recovery inhibits community transformation and creates resilience traps". In: *Nature Communications* 12.1 (2021). doi: 10.1038/s41467-021-27359-5.
- [14] N. Alemazkoor, **B. Rachunok**, D. R. Chavas, A. Staid, A. Louhghalam, R. Nateghi, and M. Tootkaboni. "Hurricane-induced power outage risk under climate change is primarily driven by the uncertainty in projections of future hurricane frequency". In: *Scientific reports* 10.1 (2020). doi: 10.1038/s41598-020-72207-z.
- [13] R. Kumar *, **B. Rachunok** *, D. Maia-Silva, and R. Nateghi. "Asymmetrical response of California electricity demand to summer-time temperature variation". In: *Scientific Reports* (2020). doi: 10.1038/s41598-020-67695-y.
- [12] A. Paulvannan Kanmani, R. Obringer, **B. Rachunok**, and R. Nateghi. "Assessing Global Environmental Sustainability Via an Unsupervised Clustering Framework". In: *Sustainability* (2020). doi: 10.3390/su12020563.
- [11] **B. Rachunok** and R. Nateghi. "The sensitivity of electric power infrastructure resilience to the spatial distribution of disaster impacts". In: *Reliability Engineering & System Safety* (2020). doi: 10.1016/j.res.2019.106658.
- [10] **B. Rachunok**, A. Staid, J.-P. Watson, and D. L. Woodruff. "Assessment of wind power scenario creation methods for stochastic power systems operations". In: *Applied Energy* 268 (2020). doi: 10.1016/j.apenergy.2020.114986.
- [9] **B. Rachunok**, J. Bennett, and R. Nateghi. "Twitter and Disasters: A Social Resilience Fingerprint". In: *IEEE Access* (2019). doi: 10.1109/ACCESS.2019.2914797.

Manuscripts Under Review & Preprints

* indicates equal contributions, Underline indicates an advisee

- [8] J. Chala Lancheros and **B. Rachunok**. "Prepare vs Response: Stochastic Modeling of Flood Resilience in NC Rural Schools". In: *Risk Analysis* (2026).
- [7] Y. Essus, R. R. Vatsavai, and **B. Rachunok**. "Data Leakage Inflates Generalizability of Power Outage Prediction Models". In: *arXiv preprint* (2026). doi: 10.48550/arXiv.2608.24665. arXiv: 2608.24665.

- [6] T. Groton and **B. Rachunok**. "Population-level measures of perceived food access reveal barriers beyond geographic proximity". In: *Nature Food* (2026). DOI: 10.48550/arXiv.2609.12132. arXiv: 2609.12132.
- [5] E. Harris and **B. Rachunok**. "Clustering-based Assessment of Power System and Energy System Inderdependence". In: *Reliability Engineering and System Safety* (2026).
- [4] R. Smith, B. McConnell, R. King, and **B. Rachunok**. "Two for One: Multi-Objective Disaster Recovery Models Improve Public Utility Post Disaster". In: *International Journal of Disaster Risk Reduction* (2026).
- [3] R. Smith, **B. Rachunok**, B. McConnell, and R. King. "Fast is Fair: Trade-offs in Cell Tower Network Restoration". In: *Reliability Engineering and System Safety* (2026).
- [2] A. Biswas, A. Helmrich, and **B. Rachunok**. "Flood-related Tweets offer High-resolution impact assessments". In: *npj Urban Sustainability* (2025).
- [1] I. Sepulveda and **B. Rachunok**. "Comparing online and offline Social Capital". In: *International Journal of Disaster Risk Reduction* (2025).

Selected Honors & Awards

KIETS Climate Leaders Program

Kenan Institute for Engineering, Technology & Science

2026

Selected with Ph.D. student Juan Chala as a 2026 Student/Mentor team.

Best Paper Award, Foundations of Risk

Society for Risk Analysis

2025

Award given to Ph.D. Student Yamil Essus's paper on ecological resilience.

C.A. Anderson Outstanding Faculty Award

NC State ISE Department

2024

Award given to one faculty per year as voted by graduating seniors

Sustainable Society Conference Delegate

University Global Partnership Network

2024

One of 8 faculty selected from NC State to attend

Rising Environmental Leadership Program Fellow

Stanford Woods Institute for the Environment

2021

Outstanding Research Award

Purdue University School of Industrial Engineering

2020

Outstanding Service Award

Purdue University School of Industrial Engineering

2020

Student Spotlight

Society for Risk Analysis Engineering and Infrastructure Specialty Group

2020

Lee Chaden Fellowship

Purdue University School of Industrial Engineering

2018

Estus and Vashti Magoon Award

Purdue University

2018

Research Advising

Current PhD Student Advisees.....

Ignacio Sepulveda

NC State University

Spring 2023-present

Industrial & Systems Engineering

Shakhawath Hossain

NC State University

Fall 2025-present

Industrial & Systems Engineering

Camryn Deames

NC State University

Co-advised with Dr. Brandon McConnell

Fall 2025-present

Business Management

Elizabeth Harris

NC State University

Fall 2025-present

Operations Research

Caroline Folsom

NC State University

Fall 2025-present

Operations Research

Juan Chala

NC State University

Fall 2025-present

Industrial & Systems Engineering

Current Master's Student Advisees.....

Zackaria Radwan

NC State University

Fall 2026-present

Operations Research

PhD Alumni.....

Yamil Essus

NC State University

Current Position: Postdoctoral Fellow, University of Toronto

Spring 2023-Spring 2026

Industrial & Systems Engineering

Robert Smith

NC State University

Co-chaired with Dr. Brandon McConnell & Dr. Russell King. Current Position: Postdoctoral Research Associate, Institute for Risk Management and Insurance Innovation, UNC-Chapel Hill

Spring 2024-Spring 2026

Operations Research

Austin Semmel

NC State University

Co-chaired with Dr. Brandon McConnell & Dr. Sebastian Heese. Current Position: U.S. Military Academy at West Point

Spring 2024-Spring 2026

Operations Research

Master's Alumni.....

Teresa Groton

NC State University

Current Position: U.S. Military Academy at West Point

Fall 2024-Spring 2026

Operations Research

Patrick Leavitt

NC State University

Current Position: Data Scientist, U.S. Coast Guard

Fall 2023-Spring 2025

Industrial & Systems Engineering

NC State Graduate Advisory Committees.....

Edric Pham

NC State University, Committee Member

Fall 2026-present

Graduate Advisory Committee

Madison Horgan

NC State University, Committee Member

Fall 2026-present

Graduate Advisory Committee

Ryan Andersen

NC State University, Committee Member

Summer 2026-present

Graduate Advisory Committee

Verónica Díaz-Pacheco

NC State University, Ph.D. Committee Member

Summer 2026-present

Operations Research

Cameron Lisy

NC State University, Ph.D. Committee Member

Summer 2026-present

Operations Research

Laura Mora

NC State University, Committee Member

Spring 2026-present

Graduate Advisory Committee

Roya Sahraei

NC State University, Committee Member

Spring 2026-present

Graduate Advisory Committee

Lindsey Jacobson

NC State University, Committee Member

Fall 2025-present

Graduate Advisory Committee

Yongseok Jeon

NC State University, Committee Member

Spring 2025-present

Graduate Advisory Committee

Lucia Rebeca Gomez Hurtado

NC State University, Committee Member

Spring 2025-present

Graduate Advisory Committee

Archit Sunil Kulkarni

NC State University, Committee Member

Spring 2025-present

Graduate Advisory Committee

Troy Wibowo

NC State University, Ph.D. Committee Member

Spring 2024-present

Industrial & Systems Engineering

Graduate School Representative Service.....

Clayson Shumway

NC State University, Graduate School Representative

Summer 2025-present
Economics & Management

External Graduate Advisory Committees.....

Cameron Moore

Duke University, Ph.D. Committee Member
Advisor: Dr. Marta Zaniolo

Spring 2026-present
Graduate Advisory Committee

Swagato Biswas Ankon

University of Georgia, MS Committee Member

Spring 2023-present
Environmental and Water Resources Engineering

Completed Graduate Advisory Committees.....

Pranav Jain

NC State University, Ph.D. Committee Member

Spring 2023-Spring 2024
Industrial & Systems Engineering

Graduate Independent Study Advisees.....

Aakash Dhruv

NC State University
Current Position: Supply Chain Engineering, Amazon

Spring 2024
Integrated Manufacturing Systems Engineering Institute

Current Undergraduate Research Advisees.....

Suyash Sapre

NC State University

2026-present
Industrial & Systems Engineering

Greta Perez

NC State University

2026-present
Industrial & Systems Engineering

Benjamin Lake

NC State University

2026-present
Industrial & Systems Engineering

Previous Undergraduate Research Advisees.....

Audrey Westlund

NC State University

Spring 2025-2026
Industrial & Systems Engineering

Ashton Sumpter

NC State University

Spring 2025-2026
Industrial & Systems Engineering

Matthew Bogdan

NC State University

Fall 2024-2026
Industrial & Systems Engineering

James Wilt

NC State University

Summer 2024-2026
Industrial & Systems Engineering REU

Ashelynn Jacobson
NC State University

Summer 2024-2026
Industrial & Systems Engineering REU

Drew Karnbad
NC State University

Spring 2024-2026
Industrial & Systems Engineering

Mike Mueller
NC State University

Spring 2024
Industrial & Systems Engineering

Victoria Mendez Gonzalez
University of Puerto Rico at Mayagüez

Summer 2023
Industrial & Systems Engineering

Fatemeh Heydari
NC State University

Summer 2023
MI-REU Program

Estevan Zamora
NC State University

Summer 2023
MI-REU Program

Julia Seidel
NC State University

Spring 2023
Honors Thesis

Adam Nayak
Stanford University

2021-2022

Policy Engagement & Government Briefings

Fayetteville City Council Work Session
North Carolina Data Center Water Resource Analysis
Invited policy briefing on data-center water, energy, wastewater, and infrastructure impacts

Fayetteville, NC
August 2026

North Carolina Energy Policy Task Force
Initial Assessment Plan: North Carolina Data Center Water Resource Analysis

Raleigh, NC
July 2026

Teaching

ISE408: *Design and Control of Production and Service Systems*

Spring 2023-Fall 2026

ISE589: *Computational Applications of Sustainability and Resilience*

Fall 2023, Fall 2024, Fall 2025, Fall 2026

Invited Seminars

ECo Talks, The Earth Commons
The Myth of Access: Affordability, Perception, and the Reality of Infrastructure Equity

Georgetown University
April 2026

CEE Graduate Colloquium
The Myth of Access: Quantifying Access to Essential Services Using AI, Durham, NC

Duke University
February 2026

Civil and Environmental Engineering
Risk, Optimization, and AI: Quantifying Access to Essential Services

Stanford University
February 2026

Environmental, Water, and Coastal Engineering Seminar Series <i>Quantifying Non-Spatial Dimensions of Access to Essential Services, Raleigh, NC</i>	NC State University <i>December 2025</i>
Society for Risk Analysis Risk Science Webinar Series <i>Quantifying and Operationalizing Resilience, Equality and Equity in Power Systems</i>	Virtual <i>April 2025</i>
AI for Extreme Weather Event and Disaster Risk Management Seminar Series <i>Quantifying Community Resilience</i>	Virtual <i>October 2024</i>
Center for Environmental and Resource Economic Policy <i>Engineering vs. Ecological Resilience</i>	NC State University <i>April 2024</i>
International Risk and Resilience Research Alliance <i>The Pitfalls of Aggregation in Assessing Community Resilience</i>	Virtual <i>February 2024</i>
Energy Data Analytics Symposium <i>Ethics & Social Responsibility of Data Analytics</i>	Duke University <i>April 2023</i>
Operations Research Seminar Series <i>Linking Sustainability & Resilience with Operations Research</i>	NC State University <i>March 2023</i>
Civil & Environmental Engineering <i>Avoiding Resilience Traps with Unsupervised Learning</i>	Cornell University <i>March 2022</i>
Industrial & Systems Engineering <i>Methods for Operationalizing Community Resilience Using Big Data</i>	NC State University <i>January 2022</i>
INFORMS Student Chapter Summer Research Series <i>Social Resilience Fingerprinting</i>	Oklahoma University <i>April 2020</i>

Invited Conference Presentations

Society for Risk Analysis Europe Annual Conference <i>Beyond Geography: Population-Scale Measurement of A-Spatial Food Access Using Large Language Models</i>	Alicante, Spain <i>May 2026</i>
Society for Risk Analysis Annual Meeting <i>Incorporating Equity and Fairness into a Mathematical Model for Power Restoration After Natural Hazards</i>	Washington, DC <i>December 2025</i>
Society for Risk Analysis Annual Meeting <i>The Future of Risk Analysis: Perspectives from Emerging Risk Professionals</i>	Washington, DC <i>December 2025</i>
INFORMS Annual Meeting <i>Incorporating Equity and Fairness into a Mathematical Model for Power Restoration After Natural Hazards</i>	Atlanta, GA <i>October 2025</i>
Society for Risk Analysis Europe Annual Conference <i>Post-disaster Power System Restoration Optimization</i>	Stavanger, Norway <i>June 2025</i>
Society for Risk Analysis Annual Meeting <i>Integrating Theories of Community Resilience</i>	Austin, TX <i>December 2024</i>

INFORMS Annual Meeting <i>Predicting and Understanding Residential Water use with Interpretable Machine Learning</i>	Phoenix, AZ October 2023
The International Symposium on Sustainable Systems and Technology <i>Quantifying Community Resilience to Break Resilience Traps</i>	Fort Collins, CO June 2023
Institute for Industrial and Systems Engineers Annual Conference <i>Quantifying Community Transformation</i>	New Orleans, LA May 2023
AGU Fall Meeting <i>Socio-hydrological impacts of water scarcity on household water affordability</i>	New Orleans, LA December 2022
Society for Risk Analysis Annual Meeting <i>Socio-hydrological impacts of water scarcity on household water affordability</i>	Tampa, FL December 2022
Society for Risk Analysis Annual Meeting <i>Quantifying Community Resilience to Break Resilience Traps</i>	Virtual December 2020
International Conference on Computational Social Science (IC2S2). <i>Be Careful Where You Get Your Tweets: Bias in Analysis from Tweet Acquisition Source</i>	Virtual June 2020
INFORMS Annual Meeting <i>Quantifying Community Transformation</i>	Virtual October 2020
Natural Hazards Center Researcher's Meeting <i>Natural Hazard Twitter Analysis Depends on Tweet Sources</i>	Virtual May 2020
INFORMS Annual Meeting <i>Social Resilience Fingerprinting</i>	Seattle, WA October 2019
Institute for Industrial and Systems Engineers Annual Conference <i>Interdependent Infrastructure System Risk & Resilience to Natural Hazards.</i>	Orlando, FL May 2019
Society for Risk Analysis Annual Meeting <i>Electric Power Infrastructure Resilience</i>	New Orleans, LA December 2018
INFORMS Annual Meeting <i>The Sensitivity of Electric Power Infrastructure Resilience to the Spatial Distribution of Disaster Impacts</i>	Phoenix, AZ October 2018
IEEE International Conference on Probabilistic Methods Applied to Power 2018 Systems <i>Stochastic Unit Commitment Performance Considering Monte Carlo Wind Power Scenarios</i>	Boise, ID June 2018
Society for Risk Analysis Annual Meeting <i>Transportation Network Recovery Analysis, an Equilibrated Perspective</i>	Crystal City, VA December 2017
INFORMS Annual Meeting <i>The Cost of Wrong Assumptions in Tropical Cyclone Frequency Distributions</i>	Houston, TX October 2017
Institute for Industrial and Systems Engineers Annual Conference <i>UAVs Provide Lifesaving Medical Care</i>	Anaheim, CA May 2016

Student Conference Presentations

Underline indicates an advisee

Society for Risk Analysis Annual Meeting

Characterization of the Dependency Between Power and Communication Outages

E. Harris and **B. Rachunok**

Washington, DC

December 2025

Society for Risk Analysis Annual Meeting

Expanding the Dimensions of Food Access: Linking Experience and Geography Through Data-Driven Analysis

T. Groton and **B. Rachunok**

Washington, DC

December 2025

93rd Military Operations Research Society Symposium

A Framework for Analyzing Operational Efficiency in US Army Aviation

Self-Organizing Map-based Clustering of Flight Dispatch Decisions

A. Semmel, H. S. Heese, B. McConnell, M. Powell, and **B. Rachunok**

Selected Best in Working Group (Readiness)

Leesburg, VA

June 2025

Society for Risk Analysis Europe Annual Conference

Quantifying Ecological Resilience

Y. Essus and **B. Rachunok**

Stavanger, Norway

June 2025

Society for Risk Analysis Annual Meeting

Fairness in Post-Disaster Power Restoration

I. Sepulveda and **B. Rachunok**

Austin, TX

December 2024

Society for Risk Analysis Annual Meeting

Quantifying Ecological Resilience

Y. Essus and **B. Rachunok**

Austin, TX

December 2024

INFORMS Annual Meeting

Fairness in Post-Disaster Power Restoration

I. Sepulveda and **B. Rachunok**

Seattle, WA

October 2024

INFORMS Annual Meeting

Estimating Access risk during blackouts due to vehicle electrification in North Carolina

Y. Essus and **B. Rachunok**

Seattle, WA

October 2024

International Symposium on Sustainable Systems and Technology

Quantifying Social Capital

I. Sepulveda and **B. Rachunok**

Baltimore, MD

June 2024

International Symposium on Sustainable Systems and Technology

Electric Vehicles Limit Equitable Access to Essential Services During Blackouts

Y. Essus and **B. Rachunok**

Baltimore, MD

June 2024

Society for Risk Analysis Annual Meeting

Quantifying Social Capital

I. Sepulveda and **B. Rachunok**

Washington, DC

December 2023

Society for Risk Analysis Annual Meeting

Electric Vehicles Limit Equitable Access to Essential Services During Blackouts

Y. Essus and **B. Rachunok**

Washington, DC

December 2023

INFORMS Annual Meeting

Electric Vehicles Limit Equitable Access to Essential Services During Blackouts
Y. Essus and B. Rachunok

Phoenix, AZ

October 2023

Scientific Community Service

Reviewer.....

Journal Articles

Communications Engineering, Journal of Infrastructure Systems, Journal of Water Resources Planning and Management, Nature Communications, IEEE Access, Risk Analysis, International Journal of Disaster Risk Reduction, Reliability Engineering and System Safety, Utility Policy, Journal of Management in Engineering

External Grant Proposals

DOE Science Synthesis Prize, November 2024

NSF Human-Environment Geographical Sciences Program, ad hoc, May 2023

NSF Human-Environment Geographical Sciences Program, ad hoc, January 2023

Conference Organizing.....

Planning Committee Member

International Symposium on Sustainable Systems and Technology (ISSST) 2026

Sustainability and Resilience of Infrastructure Track Chair

International Symposium on Sustainable Systems and Technology (ISSST) 2024

Equity, Sustainability and Engineering Session Chair

Society for Risk Analysis Annual Meeting 2023

Professional Societies.....

Society for Risk Analysis 2016-present
Council member 2021-2023; Secretary 2024-present; Board Liaison, Risk Science & Translation Committee, 2026-present

INFORMS 2017-present

Institute for Industrial and Systems Engineers 2016-present

University Service

Department and University Committees.....

ISE Masters of Engineering Management Committee Fall 2023, Spring 2024

ISE PhD Admissions Committee Spring 2023, Fall 2024, Spring 2025

Other Service.....

Engineering Open House Spring 2023, Fall 2023, Spring 2024, Fall 2024

ISE Summer Camp Summer 2023, Summer 2024